Real-Time Hand Tracking via TDMA-Based Magnetic Positioning and Machine Learning

Alex McMoil, Mingwei Li, Peng Li, *Member, IEEE*, Michael A. Urbin, and Guangyong Li, *Member, IEEE*

***Abstract*—High-fidelity hand tracking is a critical enabling technology for immersive virtual reality (VR) and intuitive human–robot interaction. While magnetic positioning (MP) offers a robust, occlusion-free alternative to optical systems, existing multinode MP architectures are often hindered by prohibitive hardware complexity, signal crosstalk, and high power consumption. This article presents a highly optimized MP system capable of simultaneous, full-hand tracking of both digits and the wrist. To address the scalability limits of conventional frequency division multiple access (FDMA) schemes, we propose a streamlined architecture based on time-division multiple access (TDMA). This design enables multiple electromagnetic transmitters to operate on a single, unified frequency driven by a localized self-oscillating circuit, thereby drastically simplifying the receiver's front-end electronics. To tackle the nonlinearities inherent in electromagnetic field modeling and anatomical variability, we employ a data-driven machine learning framework to directly regress raw magnetic flux data into precise joint angles. This approach effectively bypasses the computational overhead of analytical inverse kinematics. Experimental results demonstrate that this hybrid hardware-software co-design achieves high-precision multifinger tracking while significantly reducing both the physical form factor and the computational footprint, making it ideal for real-time wearable applications.**

***Index Terms*—Hand tracking, machine learning, magnetic positioning (MP), time-division multiplexing.**

I. Introduction

HAND gestures provide a natural and intuitive modality for communicating with computers, robots, and individuals with speech or hearing impairments. Consequently, robust hand tracking is indispensable for human–computer interaction (HCI) in virtual reality (VR), teleoperation in robotics, and automated sign language recognition. Hand tracking techniques are generally categorized into vision-based and sensor-based approaches. Vision-based tracking, both marker-based and markerless, has been successfully deployed in healthcare, driver monitoring, and device control [1]. However, these systems rely on line-of-sight and are limited by fixed camera setups, making them unsuitable for uncontrolled environments characterized by occlusion or poor lighting conditions.

To address these limitations, various sensor-based motion detection methods have been explored [2], utilizing electromechanical sensors [3], [4], [5], [6], inertial measurement units (IMUs) [7], [8], alternating current (AC)/DC magnetic flux detection [9], and electromyography (EMG) [10], [11]. Compared to vision-based techniques, sensor-based approaches offer superior flexibility and portability.

The most direct sensor-based technique involves measuring finger angles via potentiometers or encoders mounted on rigid exoskeletons [12], [13], [14], [15]. While fast and accurate, these devices are often bulky and obtrusive. Alternatively, smart gloves integrating flex sensors offer improved comfort [6], [16], [17] but typically suffer from lower accuracy and limited durability. IMUs are widely used for tracking the arm and wrist [18] but face challenges in tracking finger kinematics due to sensor size constraints [19]. Although recent innovations have embedded IMUs into smart rings [20], [21], [22], these systems remain susceptible to signal drift and are ill-suited for long-duration recording. Similarly, EMG signals from forearm muscles can estimate hand position [23] but are prone to noise, muscle fatigue, and skin conditions that affect signal quality. To achieve reliable accuracy, EMG is often fused with IMU data [22], [24], which inevitably increases system complexity.

Magnetic positioning (MP) has emerged as a robust alternative for HCI, offering high accuracy and stability. MP systems measure magnetic fields generated by permanent magnets [25] or electromagnets [26], [27]. While permanent magnets are simple, they are difficult to differentiate from one another and are susceptible to geomagnetic interference. In contrast, AC electromagnets generate fields that are easily separable from the Earth's geomagnetic field, enabling operation over longer ranges. For instance, Parizi et al. [26] demonstrated a ring-based transmitter that overcomes drift and interference, though it was limited to single-finger tracking.

To enable multifinger tracking, Santoni et al. [28] developed a multinode MP system utilizing frequency division multiple access (FDMA) [29]. While functionally effective, FDMA mandates that each transmitter node operate at a distinct

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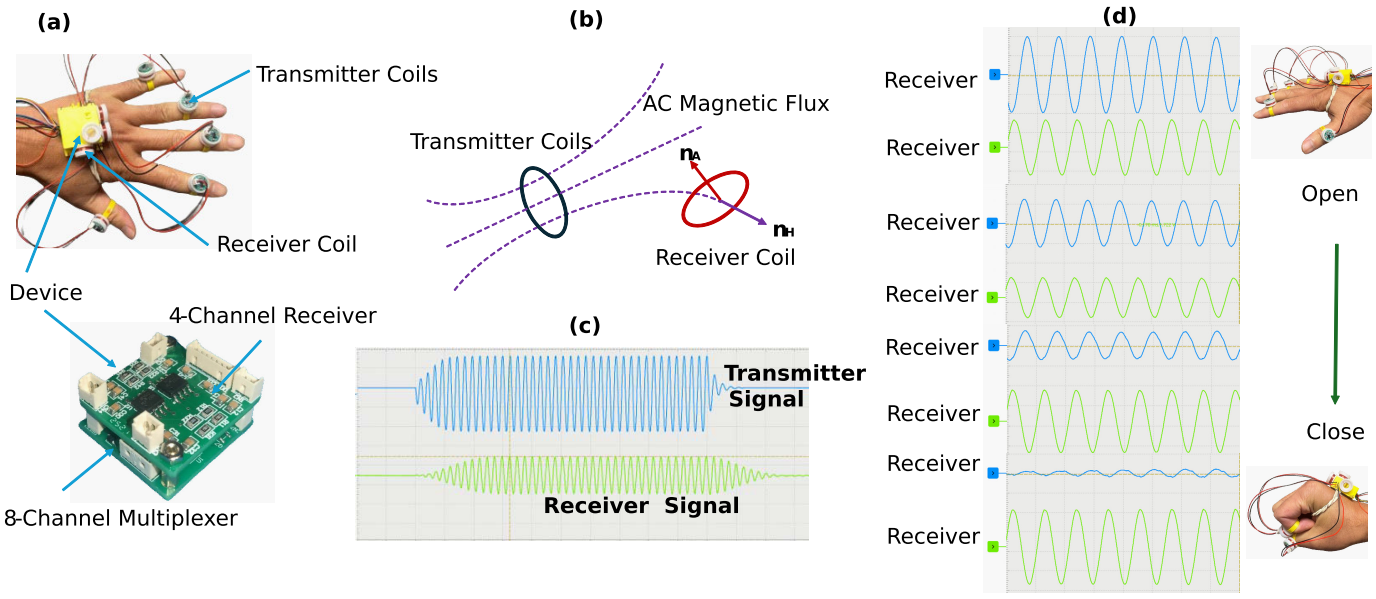


Fig. 1. Working principle of the time-division multiplexed MP system. (a) Finger movement tracking device. (b) Principle of magnetic flux detection. (c) Multiplexing channels for tracking movements of multiple fingers. (d) Receiver signal changes corresponding to index finger motion (only two receiver signals are shown for the convenience of illustration).

frequency with a specifically tuned receiver. This architectural constraint severely inflates hardware complexity, necessitating up to 24 dedicated receiver channels and multiple microcontroller units (MCUs) [30]. Consequently, these systems suffer from substantial computational overhead and increased hardware costs, ultimately diminishing their practical viability for real-time, wearable exoskeleton applications where space, power, and processing bandwidth are strictly limited.

To overcome these inherent limitations, we propose a streamlined, highly efficient architecture based on time-division multiple access (TDMA). By employing a TDMA protocol to sequentially activate transmitter nodes, our approach enables all electromagnetic sources to operate on a single, unified frequency. Consequently, this architecture allows a solitary MCU to multiplex and sample a four-channel receiver array, thereby drastically reducing both hardware complexity and computational overhead while preserving high-fidelity kinematic tracking.

The operational principle of the TDMA-enabled MP system is illustrated in Fig. 1. The physical architecture comprises six electromagnetic transmitter nodes integrated into rings worn on the digits and a four-channel receiver array secured to the dorsal aspect of the hand [Fig. 1(a)]. As the AC magnetic flux propagates [Fig. 1(b)], the TDMA protocol ensures that only a single transmitter is energized at any given instant [Fig. 1(c)]. By selectively sampling the induced signals during these discrete time slots, the system effectively isolates and resolves the specific joint angle corresponding to the active node [Fig. 1(d)]. This temporal multiplexing strategy drastically reduces the requisite number of receiver channels and MCU peripherals, all while cyclically capturing the comprehensive kinematic state of the hand and wrist. Ultimately, this architectural simplification yields a threefold advantage critical for wearable robotics: minimized manufacturing costs, significantly reduced power consumption, and a streamlined form factor that maximizes user comfort and wearability.

Finally, deriving an analytical model to map induced receiver voltages to transmitter spatial poses (position and orientation) constitutes a formidable electromagnetic inverse problem [31], [32], [33]. Moreover, translating these resolved spatial coordinates into biological joint angles necessitates complex inverse kinematic modeling, which is intrinsically confounded by subject-specific anatomical variability [25], [30]. To bypass these analytical bottlenecks, we propose a purely data-driven machine learning framework. By leveraging offline training to implicitly encode these highly nonlinear electromechanical and biomechanical relationships, our algorithm directly regresses accurate finger joint angles from raw voltage data. This approach effectively circumvents the mathematical complexities of the inverse problem and completely eliminates the need for explicit geometric or anatomical calibration.

The remainder of this article is organized as follows: Section II details the system design; Section III explains the machine learning algorithm; Section IV presents experimental validation; Section V discusses system cost and functionality as well as limitations; and Section VI concludes the study.

II. System Design

The proposed MP system is architected around four distinct functional modules, as illustrated in Fig. 2.

1) *Time-Division Multiplexed Transmitter:* A switching module that sequentially energizes individual transmitter nodes to ensure strict temporal signal isolation.
2) *Four-Channel Receiver:* A dedicated signal conditioning front-end that amplifies, rectifies, and filters the induced magnetic signals.

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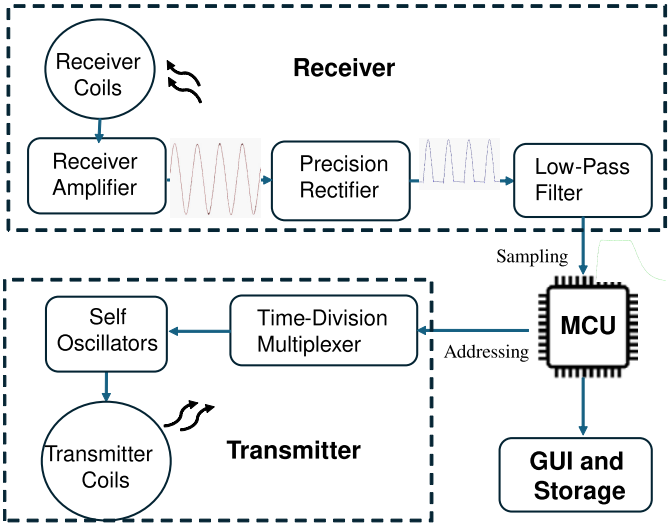


Fig. 2. Proposed MP system that consists of a time-division multiplexed transmitter with eight nodes, a group of four receivers, an MCU, and a GUI.

3) *MCU:* The central processing hub responsible for orchestrating the TDMA protocol and sampling the conditioned receiver signals.
4) *Graphical User Interface (GUI):* A comprehensive software platform on a workstation designed to execute the machine learning inference to compute real-time joint angles, manage data telemetry, and visualize real-time kinematic tracking and log telemetry data for post-hoc analysis.

In this section, we first establish the theoretical signal model governing the MP system. Subsequently, we detail the hardware architecture, providing a comprehensive analysis of the circuit designs for both the time-division multiplexed transmitter array and the signal-conditioning receiver.

A. Signal Model of the MP System

In the proposed MP system, each transmitter node consists of a coil in parallel with a capacitor, driven by an LC self-oscillating circuit. The transmitter radiates an alternating magnetic field. The magnetic flux density, $\mathbf{B}$, at a position $\mathbf{P}$ relative to the transmitter (oriented along the unit vector $\mathbf{H}$), is defined by [34]

$$\mathbf{B} = \frac{\mu_0 M_T}{4\pi}\left(\frac{3\,(\mathbf{H}\cdot\mathbf{P})\,\mathbf{P}}{R^5} - \frac{\mathbf{H}}{R^3}\right) \tag{1}$$

where μ_0 is the permeability of free space, R is the Euclidean distance from the transmitter to the receiver ($R = \|\mathbf{P}\|$), and M_T is the magnitude of the magnetic dipole moment. The dipole moment is induced by the AC current flowing through the emitter coil and can be expressed as

$$M_T = N_e A_e I \sin(\omega t) \tag{2}$$

where N_e and A_e denote the number of turns and the cross-sectional area of the emitter coil, respectively; I is the current amplitude; and ω is the angular frequency of the LC circuit. According to Faraday's law of induction, this varying magnetic flux induces a voltage at the receiver coil with a magnitude of

$$v = -N_e N_A A_e A_A I\omega\cos(\omega t)\left[\frac{3\,(\mathbf{H}\cdot\mathbf{P})\,\mathbf{P}}{R^5} - \frac{\mathbf{H}}{R^3}\right]\cdot\hat{\mathbf{n}}_{\mathbf{A}} \tag{3}$$

where N_A and A_A are the number of turns and cross-sectional area of the receiver coil, and $\hat{\mathbf{n}}_{\mathbf{A}}$ represents the orientation unit vector of the receiver coil.

As indicated by (3), retrieving the position $\mathbf{P}$ and orientation $\mathbf{H}$ from the voltage v constitutes a complex nonlinear inverse problem. Furthermore, mapping these geometric parameters to specific finger joint angles requires an inverse kinematic model, which is computationally intensive and prone to errors due to anatomical variability between subjects. To streamline analyses, we utilize machine learning algorithms to directly infer finger joint angles from raw voltage readings as part of offline training to capture these complex relationships.

B. Time-Division Multiplexed Transmitter

Implementing a centralized oscillator topology with an analog multiplexer seems a compelling approach to minimize space and component costs. However, high-frequency, high-current analog switches capable of handling oscillation signals without introducing significant distortion or insertion loss are costly. Additionally, routing high-frequency drive signals from a centralized oscillator to the distributed transmitter coils requires long transmission traces. These traces act as unintended antennas, emitting stray electromagnetic interference (EMI). In our system design, we employ localized oscillators co-located directly above each corresponding transmitter coil. This decentralized approach eliminates long, high-frequency signal routing, minimizes parasitic EMI, and ensures that each transmitter assembly behaves as an idealized, localized magnetic point source.

Each localized transmitter node employs a Colpitts oscillator, as illustrated in Fig. 3(a). This topology features an LC tank circuit—comprising two series capacitors (C_1 and C_2) in parallel with an inductor (L)—to generate stable sinusoidal oscillations upon the application of a DC voltage source (V_{in}). The resonant frequency (f) is governed by [35]

$$f = \frac{1}{2\pi\sqrt{LC_{eq}}}, \quad \text{where} \quad C_{eq} = \frac{C_1 C_2}{C_1 + C_2}. \tag{4}$$

In this specific design, a common-base amplifier configuration provides feedback from the capacitive voltage divider directly to the emitter. This topology is advantageous over a traditional common-emitter configuration because it eliminates the need for a bulky radio frequency choke between the collector and the power supply, thereby simplifying the circuit footprint. Although the common-base configuration inherently exhibits low input impedance and high output impedance, these characteristics are entirely acceptable given that the circuit functions as an isolated, self-contained oscillator rather than a signal driver.

An operating frequency of 32.5 kHz was strategically selected. This frequency is sufficiently low to remain within the gain-bandwidth product of standard low-cost operational amplifiers, yet high enough to exceed the human auditory

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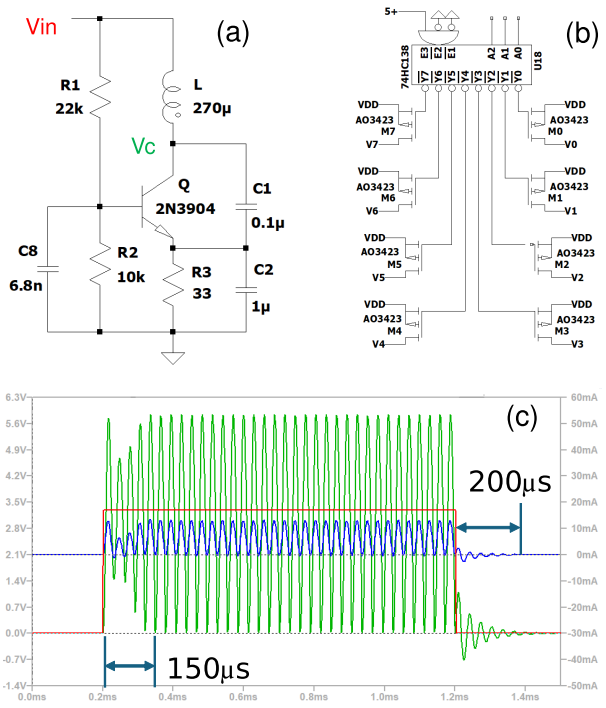


Fig. 3. Colpitts oscillator driven by a common-base amplifier and addressed by a 3–8 decoder. (a) Circuit schematic of the Colpitts oscillator. (b) Addressing logic circuit. (c) Simulation results showing the voltage supply (red) when active, the collector voltage (green), and the current through resistor R_3 (blue).

range, effectively mitigating acoustic interference from surrounding equipment. According to (4), selecting $C_1 = 0.1\,\mu$F and $C_2 = 1\,\mu$F necessitates a transmitter coil inductance (L) of approximately 270 μH. This target inductance is physically realized by winding 144 turns of 0.2 mm enameled wire around a 9 mm diameter core.

The overall architecture supports up to eight multiplexed transmitter nodes (with six actively deployed in the current prototype). This time-division multiplexing scheme is orchestrated by a 3-to-8 line decoder [Fig. 3(b)]. Simulation results [Fig. 3(c)] demonstrate that when a specific node is addressed, a 3.3 V supply is asserted for a 1 ms duration (red trace). This activation energizes the Colpitts oscillator, inducing a voltage oscillation at the transistor collector (green trace). The transient oscillation stabilizes within 150 μs of activation and fully decays within 200 μs after the supply voltage is removed. Furthermore, the circuit is highly power-efficient; drawing an average DC of merely 7 mA [blue trace in Fig. 3(c)], each node consumes less than 25 mW during its active operational phase.

C. Receiver

The receiver comprises four identical channels, each consisting of five cascading stages, as illustrated in Fig. 4(a): (I) a resonant receiver coil, (II) a pre-amplifier, (III) a precision rectifier, (IV) a passive RLC filter, and (V) a Sallen–Key low-pass filter.

Stage I (Receiver Coil): The receiver coil is paired with a parallel capacitor to form an LC resonant tank tuned to the transmitter's carrier frequency, thereby maximizing sensitivity to the magnetic induction. To facilitate robust inverse kinematic calculations, the receiver coils are spatially distributed or orthogonally aligned to minimize channel correlation.

Stage II (Pre-amplifier): A low-noise NE5532 op-amp is configured as an inverting pre-amplifier to boost the induced weak signals. This stage provides a voltage gain of 100 (40 dB) with an effective bandwidth of 100 kHz. An input coupling capacitor blocks low-frequency noise, while high-frequency interference is naturally attenuated by the op-amp's bandwidth limitations, effectively creating a bandpass response.

Stage III (Precision Rectifier): This stage employs a TL072 op-amp to convert the AC pre-amplifier output into a rectified DC envelope with an additional voltage gain of 5 (14 dB). The TL072 is selected for its high slew rate (20 V/μs), which is critical for accurately tracking the fast-rising edges of the 32.5 kHz signal during rectification.

Stage IV (RLC Filter): The rectified signal passes through a passive RLC low-pass filter optimized for fast transient response. In a TDMA system, rapid settling is essential to prevent signal bleed between sequential node activations. The resistor R_5 is sized to quickly discharge the energy stored in C_3 and L_2 during signal transitions, minimizing system lag.

Stage V (Sallen–Key Low-Pass Filter): A second-order Sallen–Key low-pass filter is implemented to attenuate residual voltage ripples. Since this stage handles near-DC signals, a general-purpose LM358 op-amp is sufficiently configured for single-supply operation to reduce power consumption and component cost.

Fig. 4(b) depicts the simulated transient response during sequential node activation (2 ms duration per node). The results indicate a settling time of approximately 0.5 ms following a switching event. Because this filter stabilization time exceeds the oscillation decay time, it serves as the dominant factor dictating the minimum permissible time slot duration (T_{slot}) per transmitter node. Theoretically, the proposed TDMA hardware inherently supports a $T_{\text{slot}} \leq 1$ ms. To ensure robust noise immunity and an adequate safety margin, we established a standardized value of $T_{\text{slot}} = 2$ ms for our system. Eight nodes will yield an overall refresh rate of 62.5 Hz [1/(8 $\times$ 2ms)], which has ample temporal resolution for tracking human finger movements. Our current prototype, configured with six transmitter nodes, gives an overall refresh rate of 83.3 Hz.

III. Hand Tracking Mechanism

While precise signal and kinematic models are conventionally considered prerequisites for MP-based tracking, this study introduces a robust, purely data-driven paradigm. Rather than deriving finger joint angles through the computationally expensive analytical inversion of induced receiver voltages, we employ a supervised machine learning framework (Fig. 5). This approach leverages offline training to map raw voltage streams directly to the kinematic parameters of a simplified hand model, implicitly decoding the complex, nonlinear electromechanical relationships.

To generate the requisite training datasets, a portable, low-cost commercial optical tracking system, Leap Motion Controller 1 (LMC1) from Ultraleap, Inc. [36], is deployed to

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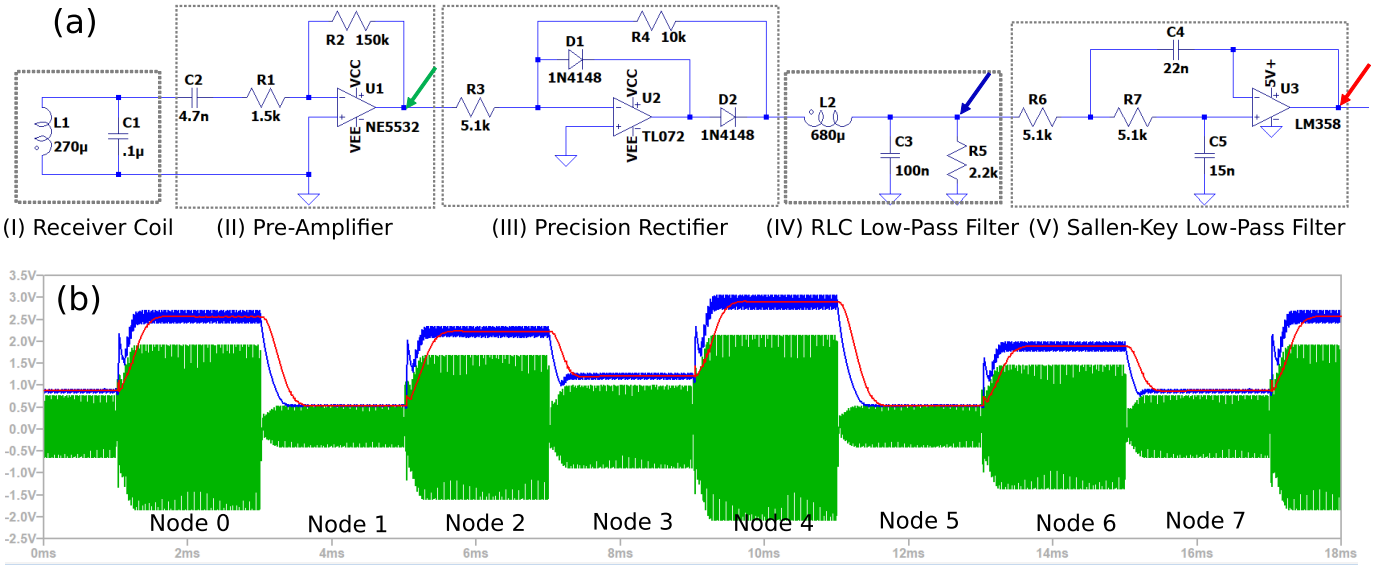


Fig. 4. Receiver: (a) circuit schematic of a single channel and (b) output of the receiver channel during sequential activation of transmitter nodes. The green, blue, and red curves indicate signals after the pre-amplifier, the RLC filter, and the Sallen–Key filter, respectively.

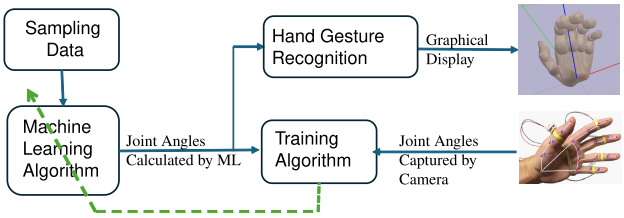


Fig. 5. MP-based hand tracking enabled by machine learning algorithms, which includes machine learning algorithms for joint angle calculation, an offline training algorithm, hand gesture recognition, and visual display.

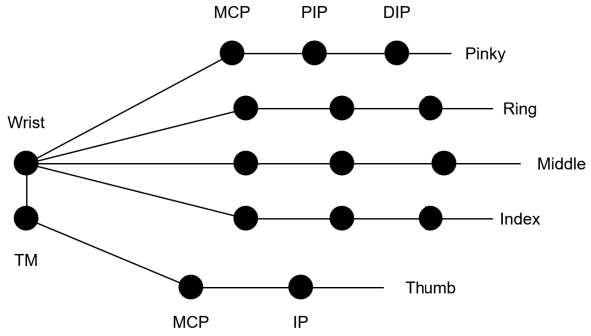


Fig. 6. Simplified kinematic model of the human hand comprising 16 DoFs: three for each digit and one for the wrist. Modified from [19].

comprehensively capture real-time, ground-truth joint angles. While the manufacturer reports a nominal hardware precision of 10 μm, practical deployment studies have characterized its operational error bounds more conservatively. Specifically, empirical literature indicates a static spatial accuracy of approximately 0.2 mm [37], which shifts to a range of 1.3–2.2 mm during dynamic hand tracking and digit articulation [38]. For the macro-gesture monitoring and tracking objectives of this study, the LMC1 provides a sufficiently precise and reliable ground-truth reference framework.

Finally, a custom-developed GUI is used to visualize predicted kinematics and rigorously validate the system's real-time tracking fidelity. The predicted joint angles drive a physics-based hand model [39] rendered within the *PyBullet* simulation environment [40], enabling continuous visual verification of the pose estimation. Furthermore, comparing these predicted kinematic profiles against the ground truth obtained from the LMC1 facilitates a rigorous evaluation of the system's tracking errors.

A. Simplified Hand Model

While the biological human hand possesses approximately 27 degrees of freedoms (DoFs) [41], rigorous dimensionality reduction is imperative to mitigate computational overhead and ensure low-latency, real-time inference. Thus, our objective is to parameterize the primary functional workspace of the hand using a minimal, yet highly representative, set of DoFs.

To establish a computationally efficient geometric foundation for our machine learning framework, we adopt a simplified kinematic hand model similar to that proposed in [19], as illustrated in Fig. 6. The wrist is mathematically abstracted as a single 1-DoF revolute joint governing flexion and extension. The thumb is parameterized with three DoFs: one addressing abduction/adduction at the trapeziometacarpal (TM) joint, and two governing flexion/extension at the metacarpophalangeal (MCP) and interphalangeal (IP) joints. The four remaining digits (index, middle, ring, and little fingers) are each allocated three DoFs, mapping directly to the flexion/extension of their respective MCP, proximal IP (PIP), and distal IP (DIP) joints.

To further condense the state space, we enforce biomechanically inspired intra-finger constraints that exploit natural tendinous coupling. Drawing upon the linear biomechanical relationships established by Hrabia et al. [42], the distal

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articulations are kinematically slaved to their proximal counterparts. Specifically, for the index, middle, ring, and little fingers, this synergistic coupling is defined as

$$\theta_{DIP} \approx 0.88\,\theta_{PIP} \tag{5}$$

and for the thumb

$$\theta_{IP} \approx 0.77\,\theta_{MCP} \tag{6}$$

where θ denotes the joint angle relative to a neutral, planar hand configuration. Consequently, while the comprehensive kinematic model explicitly maps 16 DoFs, the state vector is driven by a reduced subspace of only 11 independent parameters (one for the wrist, two for the thumb, and two for each of the four fingers). This drastic reduction critically optimizes the machine learning regression space without sacrificing grasp fidelity.

B. Machine Learning Algorithm Design

To reconstruct the comprehensive hand pose from magnetic sensor data, we implemented a supervised learning framework. To mitigate the computational burden inherent in multidimensional kinematic estimation, we decompose the hand into six independent kinematic chains: one governing the wrist and one for each of the five digits. This highly parallel, modular architecture significantly simplifies the regression problem by treating each distinct physical linkage as an isolated predictive subsystem.

1) Data Representation: The feature space is directly derived from the system's physical TDMA configuration, wherein six transmitting nodes (strategically affixed to the forearm, the thumb's proximal phalanx, and the intermediate phalanges of the fingers) are sequentially energized. Simultaneously, the four receiver coils located on the dorsal aspect of the hand quantify the induced magnetic flux densities. The input to each distinct estimator is a feature vector $\mathbf{x}_i \in \mathbb{R}^4$, representing the normalized voltage readings captured from the four receiver channels during the ith transmitter's active temporal window.

The framework aggregates these independent estimates to synthesize a global state vector $\mathbf{y} \in \mathbb{R}^{11}$. This comprehensive output array encompasses the flexion/extension and abduction/adduction angles for the wrist, the carpometacarpal joint of the thumb, and the PIP joints of digits 2 through 5. The remaining distal articulations (θ_{DIP} for the fingers and θ_{IP} for the thumb) are sequentially derived using the analytically defined intra-finger kinematic constraints from (5) and (6).

2) Joint Angle Estimation: Since the magnetic readings are highly nonlinear, as seen from (3), a dedicated Multilayer Perceptron (MLP) is deployed to regress the highly nonlinear mapping between the four-channel receiver voltage readings and the specific joint angles of each digit. For each finger, the model explicitly outputs the two independent DoFs defining its spatial configuration: flexion of the MCP and PIP joints ($\theta_{MCP}, \theta_{PIP}$). For the thumb, it resolves the abduction of the TM joint alongside the flexion of the MCP joint ($\theta_{TM}, \theta_{MCP}$). For the wrist, the model outputs the single flexion angle of the joint.

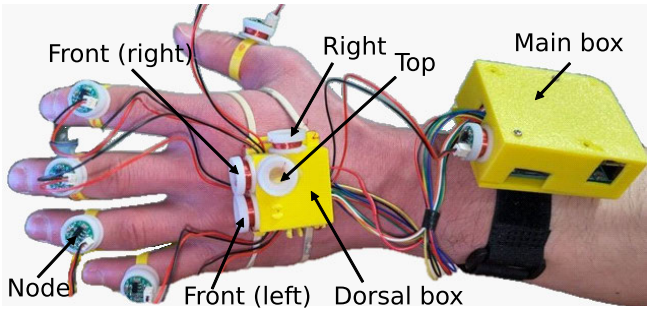


Fig. 7. Prototype of the MP-based hand tracking system.

The optimized network topology features three hidden layers, each populated with 200 neurons utilizing rectified linear unit (ReLU) activation functions to prevent gradient vanishing. The voltages are linearly scaled to a range between [0,1] to ensure convergence. Because joint angle prediction is a continuous multioutput regression problem, the network was trained using the mean squared error (mse) loss function, defined as

$$\mathcal{L}_{\text{MSE}} = \frac{1}{M \cdot D} \sum_{i=1}^{M} \sum_{j=1}^{D} \left(\theta_{i,j} - \hat{\theta}_{i,j}\right)^2 \tag{7}$$

where M is the batch size, D is the number of predicted DoFs, θ represents the optical ground-truth joint angle, and $\hat{\theta}$ represents the model's prediction.

IV. Experimental Results

Guided by the proposed system architecture and tracking algorithms, we developed a functional prototype of the MP-based hand tracking system as described in Section IV-A. The rest part of this section detail the comprehensive experimental verification and performance testing conducted to validate this prototype through a pilot study primarily involving the left hand of a healthy co-author (Subject #1, male). Additionally, the device was evaluated on the left hand of another healthy volunteer in the lab (Subject #2, male) to validate cross-user reproducibility. Unless stated otherwise, all reported experimental and testing results correspond to Subject #1 by default. As this preliminary testing was limited to noninvasive self-experimentation by the research team for system calibration, it was exempted from formal Institutional Review Board (IRB) approval.

A. Prototype Description

As illustrated in Fig. 7, the prototype system is composed of two primary modules. The first is a box housing the power supply, microcontroller, and post-amplification signal processing circuits, which can be worn on the forearm (as shown in Fig. 7) or attached to a belt strapped around the waist or placed in the user's pocket. The second is a more compact box housing the TDMA multiplexing module, receiver coils, and pre-amplifier circuits. The system features six transmitting nodes: one affixed to the first module and five ring-mounted nodes positioned on the intermediate phalanges of the fingers and the proximal phalanx of the thumb.

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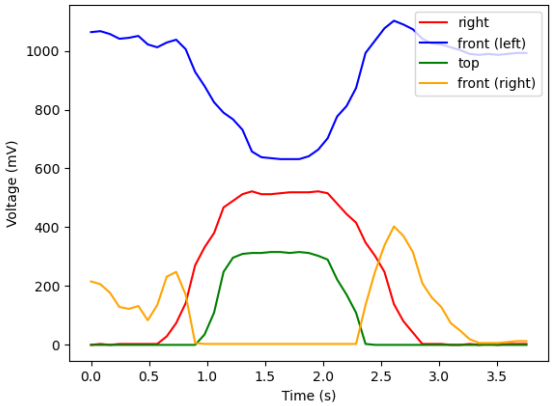


Fig. 8. Voltage profiles of the four-channel receiver for the index finger during an extension-flexion-extension movement sequence.

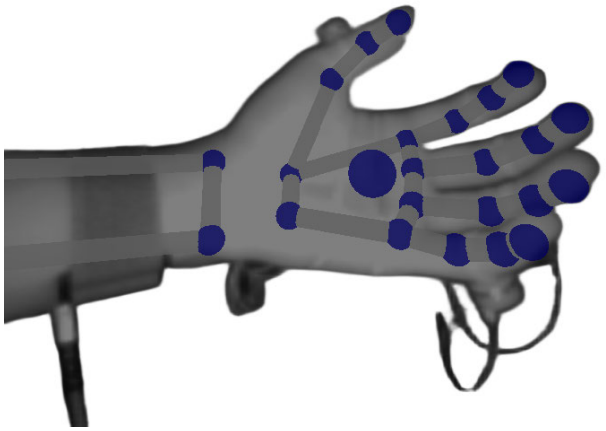


Fig. 9. Skeleton tracking visualization showing joints and bones identified by the Leap Motion Controller.

The central processing unit is an Arduino Nano 33 microcontroller, which acquires signals via the TDMA protocol. The microcontroller samples the conditioned receiver signals using its internal 10-bit analog-to-digital converters (ADCs). To ensure signal stability and accommodate transient settling times, the protocol assigns an active time interval of 2 ms to each transmitting node, performing an average of five samples per interval. With a sequence of six nodes, the system completes a complete data cycle every 12 ms, resulting in a refresh rate of approximately 83.3 Hz.

Data are subsequently streamed via a serial connection to an external workstation, where the machine learning inference is executed using the *scikit-learn* Python library. Fig. 8 depicts the raw voltage response of the four-channel receiver corresponding to the index finger during a flexion-extension cycle.

B. Dataset Collection for Training

To train the hand-tracking algorithm, we simultaneously recorded receiver voltage signals and kinematic data from the optical tracking system to serve as ground truth using the LMC1 (Ultraleap, Inc.) [36], which tracks the 3-D spatial coordinates of key hand landmarks (e.g., joints, fingertips, and forearm), as illustrated in Fig. 9.

To compute the joint angles, we model each phalanx as a vector segment. The flexion angle θ between two adjacent segments is derived from the dot product of their corresponding vectors $\mathbf{v}_1$ and $\mathbf{v}_2$

$$\theta = \arccos\left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|}\right). \tag{8}$$

Special consideration is required for the wrist to account for extension (hyperextension), which corresponds to negative angular displacement relative to the neutral position. To resolve this, we establish a palm-normal coordinate system. We define the palm normal vector by taking the cross product of the transverse vector (connecting the MCP joints of the index and ring fingers) and the longitudinal vector (representing the middle finger's metacarpal). The wrist angle is subsequently calculated as the angle between this palm normal vector and the forearm vector, offset by $-\pi/2$ to normalize the reading to a flat-hand resting pose.

A comprehensive dataset consisting of a sufficient number of data frames was captured from the subject's left hand during continuous gesture cycles. The system architecture is inherently subject-specific, accommodating variations in individual hand dimensions and sensor placement geometries. Due to differences in range of motion (RoM), data for the thumb, wrist, and remaining fingers were recorded in separate sessions. We captured approximately one minute of continuous data (1800 samples at 30 Hz) for both the thumb and nonthumb fingers, while the relative kinematic simplicity of the wrist required only 20 s (600 samples). In all sessions, data points were collected and divided into independent subsets: training (70 %), validation (10 %), and testing (20 %). Model training was executed using the Adam optimizer to minimize the mse loss function, configured with a batch size of 200 and an initial learning rate of 10^{-3}. To prevent model overfitting and ensure smooth decision boundaries across the nonlinear electromagnetic fields, an early stopping mechanism with a patience threshold of ten epochs and an L_2 regularization penalty ($\lambda = 10^{-4}$) was incorporated into the training loop.

C. Experimental Validation

Following the structured acquisition of the subject-specific dataset (Section IV-B), we evaluated the intra-subject tracking fidelity of our custom regression models (Section III-B) at different motion speeds: slow (around 10 s), medium (around 3 s), and fast (around 1 s), for a full cycle of motions. Kinematic performance was evaluated by benchmarking joint angles predicted by our machine learning framework against ground-truth data synchronized via the leap motion protocol.

Fig. 10 illustrates the real-time tracking results for the index finger during a complete flexion-extension cycle with a slow-to-medium motion speed. The red curves denote the predicted angles for the MCP [Fig. 10(a)] and PIP [Fig. 10(b)] joints, while the blue curves represent the ground truth. The system demonstrates robust tracking capabilities with minimal lag. The tracking performance for other digits is very similar, so they are not presented for conciseness.

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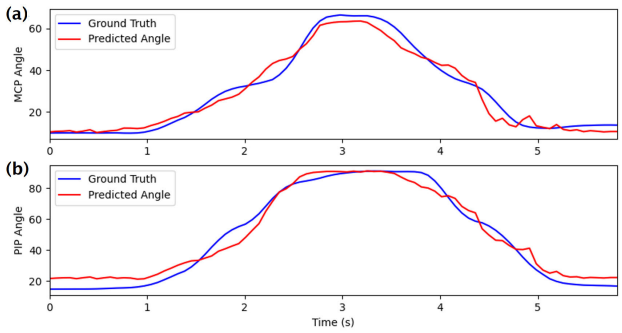


Fig. 10. Tracking performance for the index finger: (a) MCP joint and (b) PIP joint angles during a flexion-extension cycle in about 4 s.

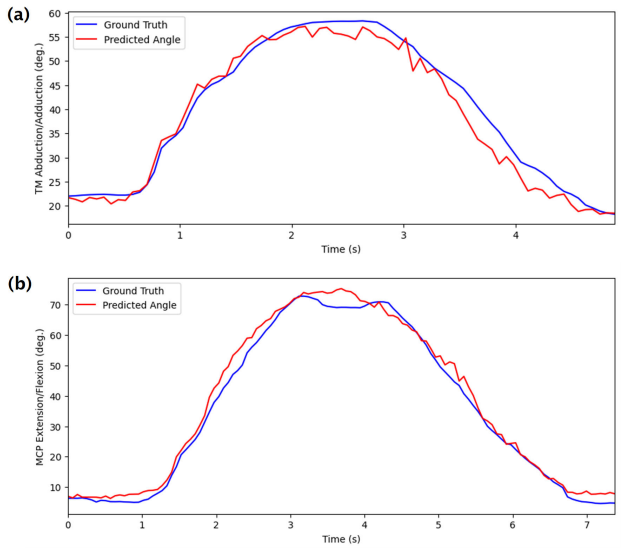


Fig. 11. Tracking performance for the thumb: (a) TM joint during adduction/abduction in about 4 s and (b) MCP joint during flexion/extension in about 6 s.

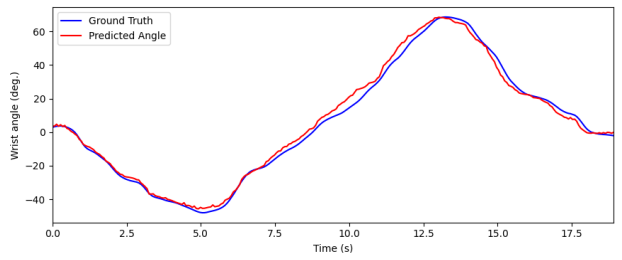


Fig. 12. Tracking performance for the wrist during an extension-flexion-extension sequence in about 18 s.

The performance of thumb tracking is presented in Fig. 11. Fig. 11(a) displays the abduction/adduction tracking of the TM joint at medium speed while the MCP joint is held at a fixed angle. In contrast, Fig. 11(b) shows the flexion/extension tracking of the MCP joint at slow speed while the TM joint remains fixed.

Fig. 12 depicts the wrist tracking performance during a slow-speed dynamic sequence: extending upward from neutral, flexing downward, and returning to neutral. The predicted angle (red) closely follows the ground truth (blue) throughout the motion.

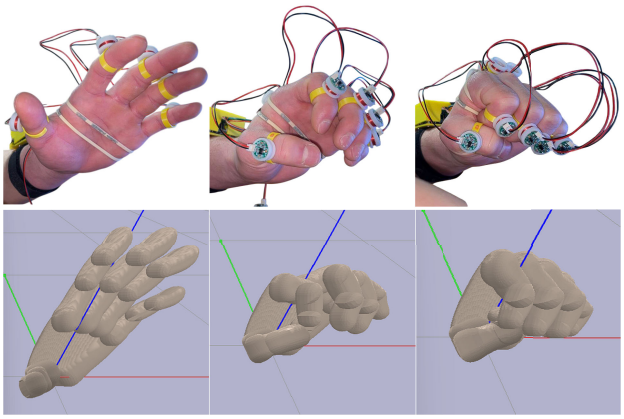


Fig. 13. Qualitative evaluation: side-by-side comparison of the real-world hand and the PyBullet-simulated hand model driven by the MP system.

TABLE I
TRACKING ERRORS OF THE SYSTEM FOR SUBJECT #1

Digit	Joint	Mean Absolute Error (deg.) Testing Dataset	Slow	Medium	Fast
Thumb	TM	2.14	4.41	5.39	6.46
	MCP	2.21	2.99	3.20	3.31
Index	MCP	3.84	3.47	4.19	6.22
	PIP	4.02	4.87	4.76	7.17
Middle	MCP	4.21	3.43	4.70	8.04
	PIP	5.04	4.21	6.46	10.56
Ring	MCP	5.20	3.93	5.94	10.05
	PIP	4.10	3.83	5.27	8.63
Pinky	MCP	4.40	3.75	5.63	8.84
	PIP	3.47	3.23	4.96	8.00
Wrist	F/E	4.13	4.45	6.44	10.93

To provide a qualitative visual evaluation, Fig. 13 presents a side-by-side comparison of the physical hand and a virtual twin that is rendered in *PyBullet*, as stated in Section III, in which the predicted joint angles drive a physics-based hand model. Clearly, the motion of the physics-based hand model tracks the real hand motions without visual discrepancy.

To quantitatively evaluate the prototype's tracking performance, tracking errors were calculated relative to the ground truth data acquired from the LMC1. These results are summarized in Table I, which details the mean absolute error (MAE) in degrees across the entire evaluation dataset, categorized by movement speeds: slow (cycle duration 10 s), medium (cycle duration 3 s), and fast (cycle duration 1 s). To thoroughly assess the system's operational RoM, distinct kinematic sequences were executed for the thumb, wrist, and nonthumb digits. The testing protocol for the nonthumb digits consisted of a continuous extension–flexion–extension sequence. The wrist followed a similar extension–flexion–extension trajectory, expanded to incorporate full hyperextension. For the thumb, the TM and MCP joints were articulated through a diverse range of angular configurations by performing a continuous circular motion (circumduction). To ensure the

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reliability and repeatability of the recorded metrics, the experimental motion sequences for the wrist and all digits were repeated for at least ten trials.

The results presented in Table I indicate that the tracking errors under slow-speed, real-world scenarios are only marginally larger or smaller than those obtained from the offline validation dataset. This high degree of consistency suggests that the MLP architecture is fully capable of capturing the severe nonlinearities inherent to the inverse magnetic mapping. Remarkably, the tracking errors only increase slightly as the execution speed accelerates from slow to medium, indicating the dynamic robustness of the device. Conversely, the system exhibits a more pronounced degradation at elevated operational velocities; however, this remains within acceptable limits as such rapid movements seldom occur during typical user operations.

Overall, our results demonstrate that the predicted and ground-truth joint angles are strongly correlated, with the majority of MAE values falling between 3° and 6° at slow and medium motion speeds. These metrics align closely with the state of the art, such as the tracking systems proposed by Parizi et al. [26] (which reported a 3.63° error using an iterative model and 4.72° using a neural network framework for general orientation) and Santoni et al. [30] (which yielded an average joint error of 5.5° for both the ring finger and the thumb). The observable discrepancies in our outcomes are primarily attributed to localized limitations in the ground-truth acquisition pipeline—specifically the depth ambiguity and spatial occlusion inherent in optical tracking systems—rather than inaccuracies within the magnetic tracking device itself. Future iterations of this architecture would benefit from next-generation depth-sensing hardware (e.g., the Leap Motion Controller 2) to improve baseline calibration accuracy. Furthermore, expanding the underlying kinematic model to incorporate additional DoFs will further enhance the tracking fidelity.

TABLE II
TRACKING ERRORS OF THE SYSTEM FOR SUBJECT #2

Digit	Joint	Mean Absolute Error (deg.) Testing Dataset	Slow	Medium	Fast
Thumb	TM	1.24	2.88	3.80	5.35
	MCP	2.16	2.52	4.67	5.49
Index	MCP	2.27	5.44	3.67	5.03
	PIP	3.15	3.97	6.07	7.81
Middle	MCP	3.46	4.38	4.03	6.38
	PIP	4.19	4.33	5.08	6.82
Ring	MCP	3.53	3.94	4.83	6.41
	PIP	4.12	5.17	5.83	7.58
Pinky	MCP	3.71	3.65	6.37	8.45
	PIP	2.56	4.16	5.14	6.76
Wrist	F/E	4.48	6.11	7.89	8.97

V. DISCUSSION

It should be noted that the proposed system is inherently subject-specific due to anthropometric variations in individual hand geometry and size. Consequently, each user requires a one-time initialization (calibration) phase before the device can be used in real-world applications. Provided that the wearable architecture is consistently positioned at the same approximate anatomical location, no subsequent recalibration or retraining is required for future use by the same user. To validate the cross-user reproducibility and robustness of the tracking device, tracking error metrics were similarly evaluated for a second subject (Subject #2) following an identical experimental protocol to that of Subject #1. The results, compiled in Table II, demonstrate comparable error distributions to those presented in Table I. This strong agreement successfully validates the cross-user reproducibility and robustness of the tracking device.

While a neural network architecture was selected as the primary regression model for this system, alternative machine learning algorithms were also investigated. For instance, a baseline Decision Tree regressor was prototyped and demonstrated tracking performance comparable to that of the neural network within the strict boundaries of the training dataset. However, because decision trees utilize discontinuous, axis-aligned step functions to partition the feature space, they exhibit excessively high sample complexity when approximating smooth, continuous physical manifolds, such as the inverse $1/R^3$ and $\mathbf{B}\cdot\hat{\mathbf{n}}_{\mathbf{A}}$ electromagnetic profiles inherent to this system. Consequently, achieving equivalent tracking precision with a decision tree mandates a significantly larger training dataset, which heavily increases the user calibration overhead. Furthermore, polynomial regression techniques cannot yield a reliable inverse mapping due to the extreme nonlinearity between the measured magnetic flux density B and the target finger joint angles θ_{MCP} and θ_{PIP}. Low-order polynomials underfit the system dynamics by failing to capture severe localized gradient changes, whereas high-order polynomials exhibit severe oscillatory behavior between exact training coordinates (Runge's phenomenon), drastically impeding model generalization.

Frequency stability of the transmitters and its impact on system performance can be categorized into two distinct effects: static component mismatch and dynamic environmental drift. Variations due to manufacturing tolerances (e.g., slight discrepancies in nominal capacitance and inductance values) result in fixed, time-invariant frequency offsets between the transmitters. Because these offsets are constant, their geometric and electromagnetic consequences are implicitly captured and accommodated by the data-driven machine learning framework during the offline training phase. Consequently, static component variations do not degrade tracking performance. Conversely, dynamic frequency shifts induced by external operating conditions—such as the close proximity of opportunistic ferromagnetic materials—alter the inductance of the transmitter coils in real time. Because this dynamic drift is nonconstant and environment-dependent, it cannot be fully compensated by a static training dataset.

The current prototype employs a distributed architecture comprising primary and secondary modules, as well as distal transmitter nodes, connected through wires. While this multicomponent design has demonstrated efficacy in signal

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acquisition, the form factor imposes constraints on wearability. Future iterations will target the integration of the primary and secondary modules into a unified, wrist-mounted form factor. Alternatively, the distal transmitter nodes can be decoupled via wireless connectivity. This wireless link may be realized either over a single frequency channel utilizing paired encoder/decoder integrated circuits (ICs, e.g., HT12A/HT12D) or across multiple channels using tone decoders (e.g., LM567) to achieve frequency-division multiplexing (FDM). However, this wireless approach necessitates independent power sources and auxiliary management circuitry for each node, which inevitably increases the volumetric footprint and overall form factor of the transmitter modules. Furthermore, transitioning to a wireless architecture introduces significant challenges regarding network synchronization and deterministic data scheduling. Specifically, implementing a TDMA framework requires precise, low-overhead encoding and decoding protocols to maintain strict temporal synchronization between individual transmitter node activations and the corresponding receiver sampling intervals.

In terms of kinematic modeling, the current system excludes finger abduction (except the thumb) and wrist deviation (radial/ulnar) due to limitations in the ground truth calibration pipeline. Future work will leverage next-generation optical tracking hardware, such as the Leap Motion Controller 2, which offers improved dynamic precision (0.4–0.8 mm) [38] to acquire high-fidelity ground truth data for these lateral movements. This will allow us to extend the model to capture a fuller range of hand DoFs.

Furthermore, while the hardware supports eight transmitter channels, only six are currently active. We intend to utilize the two dormant nodes, allocating one to the thumb and one to the wrist to further enhance tracking resolution. The system can be expanded to 16 transmitter nodes by cascading an additional 3-to-8 line decoder. This hardware expansion modifies the overall system refresh rate to 31.25 Hz, which is still sufficient for most real-world, human–machine interface applications. For bimanual tracking scenarios requiring simultaneous monitoring of both hands, each independent hand module can be configured to operate at a distinct, orthogonal frequency. This hybrid multiplexing approach effectively eliminates inter-system crosstalk between the two sides.

Finally, the system's reliance on a stationary workstation for machine learning inference currently constrains mobility. To achieve a truly portable, standalone solution, future development will focus on migrating the computational load to edge computing platforms, such as mobile devices or single-board computers (e.g., Raspberry Pi).

VI. Conclusion

In this article, we present a robust and cost-effective framework for hand tracking based on MP. The proposed system features a distributed hardware architecture that utilizes self-oscillating Colpitts circuits to tune the electromagnetic transmitter nodes to a common frequency. To manage signal acquisition, we implemented a TDMA protocol that sequentially activates the nodes, coordinated by a microcontroller and a four-channel receiver for the precise detection of magnetic flux. Compared to previous work using the FDMA protocol, our TDMA approach significantly reduces the complexity of the system. This reduction directly translates into lower manufacturing costs, decreased power consumption, and improved wearability due to a more compact form factor.

A further key contribution of this study is the shift from traditional analytical modeling to an automated, machine learning approach. Rather than solving the computationally expensive and often ill-posed inverse magnetic problem, we demonstrate that finger joint angles can be directly estimated using machine learning algorithms trained on a simplified kinematic model. Experimental results validate the efficacy of this framework, demonstrating that predicted joint angles are nearly identical to those derived from the optical tracking system.

Future work will focus on improving tracking fidelity by integrating next-generation ground-truth hardware (e.g., Leap Motion Controller 2) to accurately capture lateral finger movements and wrist deviations, thereby facilitating ubiquitous applications in VR, augmented interfaces, and telerobotic control.

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